2(a)	a = gradient or a = Δv / (Δ)t or a = (v – u) / t a = (0.30 – 0.12) / (0.35 – 0.15) a = 0.90 m s^{-2}	[1] [1]
2(b)	By conservation of momentum, $P_i = P_f$ $\rightarrow$: (0.25 × 0.48) + (0.75 × 0.12) = (0.25 v) + (0.75 × 0.30) or Relative speed of approach = relative speed of separation $\rightarrow$: $u_A - u_B = v_B - v_A$ 0.48 – 0.12) = (0.30 – v) v = – 0.060 m s^{-1} or $KE_i = KE_f$ $(1/2 \times 0.25 \times 0.48^2) + (1/2 \times 0.75 \times 0.12^2) = (1/2 \times 0.25 \times v^2) + (1/2 \times 0.75 \times 0.30^2)$ v = 0.060 m s^{-1} Magnitude of v: 0.060 m s^{-1} Direction: opposite to initial velocity of A / opposite to initial /final velocity of B/ to the left	[1] [1] [1]